Lecture 2

Recap:

1. The complex Fourier series of a T-periodic f is given by

$$f(t) \sim \sum_{n=-\infty}^{\infty} c_n e^{in\Omega t}, \quad \Omega = \frac{2\pi}{T}$$

$$c_n = \frac{1}{T} \int_0^T f(t) e^{-in\Omega t} \, dt$$

- Can shift the range of integration $\rightarrow \int_a^{a+T}$
- Choose the easiest range for the problem.

2. The sawtooth wave has Fourier series

$$f(t) \sim \sum_{n \neq 0} \frac{(-1)^{n+1}}{in} e^{int}$$

1.3 Trig. form of Fourier series

Can use Euler's formula to write the series as

$$f(t) \sim \sum_{n=-\infty}^{\infty} \left(c_n \cos n\Omega t + i c_n \sin n\Omega t \right).$$

Now split up the sum as follows

$$f(t) \sim \left(\sum_{n=-\infty}^{-1} c_n \cos n\Omega t + i c_n \sin n\Omega t \right) + c_0 + \left(\sum_{n=1}^{\infty} c_n \cos n\Omega t + i c_n \sin n\Omega t \right)$$

In the first sum we will redefine $n \mapsto -n$ and use $\cos(-x) = \cos(x)$ & $\sin(-x) = -\sin(x)$ to obtain

$$f(t) \sim c_0 + \sum_{n=1}^{\infty} \Big[(c_n + c_{-n}) \cos(n\Omega t) + i(c_n - c_{-n}) \sin(n\Omega t) \Big]$$

$$\doteq \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos n\Omega t + b_n \sin n\Omega t \right)$$

where

$$a_n = c_n + c_{-n}, \quad b_n = i(c_n - c_{-n}).$$

Using the integral expression for c_n:

$$a_n = \underbrace{\frac{1}{T} \int_0^T f(t) e^{-in\Omega t} dt}_{c_n} + \underbrace{\frac{1}{T} \int_0^T f(t) e^{in\Omega t} dt}_{c_{-n}}$$

$$\Rightarrow a_n = \frac{1}{T}\int_0^T f(t)\left(e^{in\Omega t} + e^{-in\Omega t}\right)dt$$

$$= \frac{2}{T}\int_0^T f(t)\cos n\Omega t \; dt.$$

Similarly, b_n is given by

$$b_n = \frac{i}{T}\int_0^T f(t)\,e^{-in\Omega t}\,dt - \frac{i}{T}\int_0^T f(t)e^{in\Omega t}dt$$

$$= \frac{1}{T}\int_0^T f(t)\, i\left(e^{-in\Omega t} - e^{in\Omega t}\right)dt$$

$$= \frac{2}{T}\int_0^T f(t)\,(-i)\sin(n\Omega t)\,dt$$

$$= \frac{2}{T}\int_0^{\frac{T}{2}} f(t)\sin n\Omega t \; dt.$$

Example 1.5 The complex coeffs. for the sawtooth are

$$c_n = (-1)^{n+1}/in \quad , \quad n \neq 0.$$

Therefore,

$$a_n = c_n + c_{-n} = \frac{(-1)^{+n+1}}{in} + \frac{(-1)^{-n+1}}{-in}$$

$$= 0 \qquad \left((-1)^n = (-1)^{-n}\right)$$

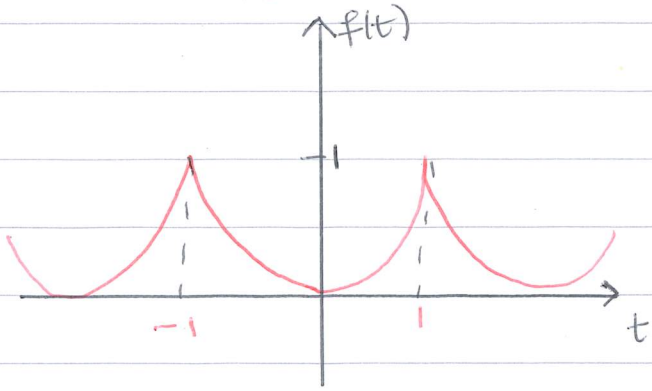
$$b_n = i(c_n - c_{-n}) = i\left(\frac{(-1)^{+n+1}}{in} - \frac{(-1)^{-n+1}}{-in}\right)$$

$$= 2(-1)^{n+1}/n$$

Extra example

Consider the 2-periodic f^n defined on $[-1,1)$ by

$$f(t) = t^2.$$



$\rightsquigarrow T = 2, \Omega = \pi$

The complex Fourier coefficients are

$$c_n = \frac{1}{2}\int_{-1}^{1} t^2 e^{-in\pi t}\,dt$$

$$= -\frac{1}{2}\left[\frac{t^2 e^{-in\pi t}}{in\pi}\right]_{-1}^{1} + \frac{1}{2\pi in}\int_{-1}^{1} 2te^{-in\pi t}\,dt$$

$n \neq 0$

$$= \frac{-1}{2\pi in}\left(e^{-in\pi} - e^{in\pi}\right) - \frac{1}{(\pi in)^2}\left[te^{-in\pi t}\right]_{-1}^{1}$$

$$+ \frac{1}{(\pi in)^2}\int_{-1}^{1} e^{-in\pi t}\,dt$$

Seen this before! Equals 0.

$$= -\frac{1}{2\pi in}\left((-1)^{-n} - (-1)^{n}\right) - \frac{1}{(\pi in)^2}\left(e^{-in\pi} + e^{-in\pi}\right)$$

$(-1)^{-n} = (-1)^{n}$

$$= +\frac{1}{\pi^2 n^2}\left((-1)^{n} + (-1)^{-n}\right)$$

$$= \frac{2(-1)^{n}}{\pi^2 n^2}$$

Now taking $n=0$

$$c_0 = \frac{1}{2}\int_{-1}^{1} t^2 dt = \frac{1}{6}\left[t^3\right]_{-1}^{1} = \frac{1}{3}$$

Therefore,

$$f(t) \sim \frac{1}{3} + \sum_{n\neq 0} \frac{2(-1)^n}{\pi^2 n^2} e^{in\pi t}$$

Trig coefficients

1) $$a_n = c_n + c_{-n} = \begin{cases} 2/3 & , n=0 \\ \frac{4(-1)^n}{\pi^2 n^2} & , n\neq 0 \end{cases}$$

2) $$b_n = i(c_n - c_{-n}) = \frac{2(-1)^n}{\pi^2 n^2} - \frac{2(-1)^{-n}}{\pi^2(-n)^2}$$

$$= \frac{2(-1)^n}{\pi^2 n^2} - \frac{2(-1)^n}{\pi^2 n^2}$$

$$= 0.$$

$$\Rightarrow f(t) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos n\pi t + b_n \sin n\pi t)$$

$$= \frac{1}{3} + \sum_{n=1}^{\infty} \frac{4(-1)^n}{\pi^2 n^2} \cos n\pi t.$$

As an exercise, calculate a_n & b_n using the integral expressions for the two examples above.

In these examples, one of a_n or b_n vanished. This is due to the 'parity' of $f(t)$

Def 1.3: A fⁿ $f: \mathbb{R} \to \mathbb{C}$ is even if

$$f(-t) = f(t), \quad \forall t \in \mathbb{R}$$

and odd if

$$f(-t) = -f(t), \quad \forall t \in \mathbb{R}.$$

Even	Odd
$\cos t$	$\sin t$
t^2	t
t^4	t^3
$\lvert t \rvert$	$\operatorname{sign}(t)$

Note: Since $e^{it} = \cos t + i \sin t$, it is neither odd nor even.

Consider some function $f: \mathbb{R} \to \mathbb{C}$ & take its integral over $[-a, a]$ $(a > 0)$

$$\int_{-a}^{a} f(t)\,dt = \int_{-a}^{0} f(t)\,dt + \int_{0}^{a} f(t)\,dt$$

$$= \int_{0}^{a} f(-s)(-ds) + \int_{0}^{a} f(t)\,dt$$

Sub in $s = -t$. ⟶

The minus sign switches the limits of the first integral. Therefore,

$$\int_{-a}^{a} f(t)\,dt = \int_{0}^{a} (f(t) + f(-t))\,dt.$$

If f is odd or even, we have a simplification

$$\int_{-a}^{a} f(t)\,dt = \begin{cases} 2\int_0^a f(t)\,dt, & \text{even} \\ 0, & \text{odd.} \end{cases}$$

Why this is useful: If we integrate our Fourier coeffs. over $[-T/2, T/2)$ we often get simplifications like above if f is even/odd!

Note: If f is odd & g is even, then $f(t)g(t)$ is odd. Similarly, if both are odd then $f(t)g(t)$ is even. We have the multi. table on the right

g \ f	odd	even
odd	even	odd
even	odd	even

Returning to the sawtooth, $f(t) = t$ is odd over $[-\pi, \pi)$

$$\Rightarrow f(t)\cos nt \text{ is odd}$$
$$\Rightarrow \int_{-\pi}^{\pi} f(t)\cos nt\,dt = 0,$$

so only the b_n terms survive!

Example 1.6 The function $f(t) = |\sin t|$ is even and has period π.

$\rightsquigarrow T = \pi$, $\Omega = 2$.

Since it is even, $f(t)\sin 2nt$ is odd

$$\Rightarrow b_n = \frac{2}{\pi}\int_{-\pi/2}^{\pi/2} f(t)\sin 2nt\,dt = 0.$$

We only need to find a_n!

$$a_n = \frac{2}{\pi}\int_{-\pi/2}^{\pi/2} |\sin t|\cos 2nt\,dt$$

$$= \frac{4}{\pi}\int_{0}^{\pi/2} \sin t\cos 2nt\,dt$$

↓ formula sheet!

$$= \frac{2}{\pi}\int_{0}^{\pi/2} \left(\sin(1+2n)t + \sin(1-2n)t\right)dt$$

$$= \frac{2}{\pi}\left[-\frac{\cos(1+2n)t}{1+2n} - \frac{\cos(1-2n)t}{1-2n}\right]_0^{\pi/2}$$

$\cos\left[(1+2n)\frac{\pi}{2}\right] = \cos\left[(1-2n)\frac{\pi}{2}\right] \neq 0$

$$= \frac{2}{\pi}\left[\frac{1}{1+2n} + \frac{1}{1-2n}\right] = \frac{4}{\pi(1-4n^2)}$$

Note that $n=0$ is not a pole, so our expression holds for $n=0$ also. Hence

$$f(t) \sim \frac{2}{\pi} - \frac{4}{\pi}\sum_{n=1}^{\infty} \frac{\cos 2nt}{4n^2-1}.$$

1.4 Half range series

Suppose we want to find a Fourier series for a f^n defined on a range $[0, T/2]$, then we can extend it to a periodic f^n over all of $\mathbb{R}$. There are two nice ways of doing this.

1) The even extension F_{even}:

$$F_{even}(t) = \begin{cases} f(t), & 0 \leq t < T/2, \\ f(-t), & -T/2 \leq t < 0. \end{cases}$$

Then we extend periodically. Since this is even, the Fourier series is

$$F_{even} \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(n\Omega t)$$

with $\Omega = 2\pi/T$ and

$$a_n = \frac{4}{T} \int_0^{T/2} f(t) \cos(n\Omega t)\, dt.$$

We call this the cosine series of f.

2) The odd extension F_{odd}:

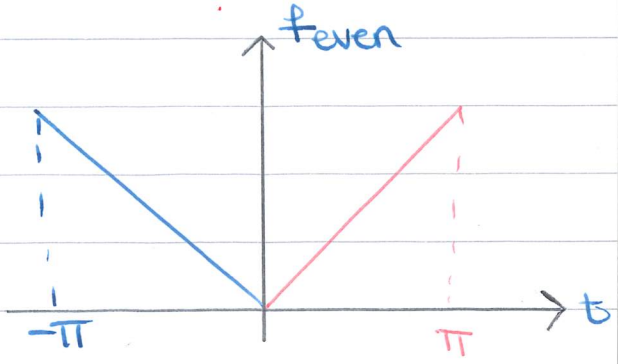
$$F_{odd}(t) = \begin{cases} f(t), & 0 \leq t < T/2 \\ -f(-t), & -T/2 \leq t < 0. \end{cases}$$

$$\Rightarrow F_{odd} \sim \sum_{n=1}^{\infty} b_n \sin n\Omega t$$ ← sine series of f.

$$\text{with } b_n = \frac{4}{T} \int_0^{T/2} f(t) \sin(n\Omega t)\, dt$$

Example 1.7

Suppose $f(t) = t$ on $[0, \pi)$. The odd extension is the sawtooth, so lets look at the even extension:



$$F_{even}(t) = \begin{cases} t, & 0 \leq t < \pi \\ -t, & -\pi \leq t < 0. \end{cases}$$

The cosine coefficients (taking $T/2 = \pi$) are

$$a_n = \frac{4}{(2\pi)} \int_0^{\pi} t \cos nt \, dt$$

$$= \frac{2}{\pi} \left[\frac{t \sin nt}{n} \right]_0^{\pi} - \frac{2}{\pi n} \int_0^{\pi} \sin nt \, dt$$

$\sin n\pi = \sin 0 = 0$

$$= \frac{2}{\pi n^2} \left[\cos nt \right]_0^{\pi}$$

$n \neq 0$

$$= \frac{2}{\pi n^2} \left(\cos n\pi - 1 \right) = \frac{2}{\pi n^2} \left((-1)^n - 1 \right).$$

Taking $n = 0$, we have

$$a_0 = \frac{2}{\pi} \int_0^{\pi} t \, dt = \frac{2}{\pi} \left[\frac{t^2}{2} \right]_0^{\pi} = \pi.$$

$$\Rightarrow F_{even}(t) \sim \frac{\pi}{2} + \sum_{n=1}^{\infty} \underbrace{\frac{2\left((-1)^n - 1\right)}{\pi n^2}}_{0 \text{ when } n \text{ is even}} \cos nt.$$